

This video elucidates the core mathematical concepts and practical applications of random variables, focusing on **Expected Value (Mean), Variance, and Standard Deviation**.

Summary

The analysis of random variables aims to reduce complex datasets into meaningful representative values.

- **Expected Value (Mean)** acts as the *Center of Mass* or balance point of a distribution.
- It reflects the **long-term average outcome** over many trials.
- A high probability of winning in a single trial does not imply a favorable long-term outcome.
- **Variance** and **Standard Deviation** measure dispersion — the spread of data around the mean.
- These concepts are foundational in statistics, physics, data science, and machine learning.

1. Concept of Expected Value (Mean)

Expected Value is denoted as $E[X]$ or μ .

It represents the center value of a distribution.

1.1 Theoretical Framework

- **Balance Point:** The weighted distances of all data points sum to zero.
- **Representativeness:** A single average represents the performance of an entire dataset.
- **Long-Term Average:** Reflects behavior over large n trials rather than a single outcome.

1.2 Mean vs Median

- **Mean:** Center of mass / balance point.
- **Median:** Literal mid-value of a dataset.

2. Mathematical Formulations

The formulas differ for discrete and continuous random variables.

Table 1: Formula Comparison

Property	Discrete Random Variable	Continuous Random Variable
Data Type	Integers / Distinct values	Defined over an interval
Expected Value $E[X]$	$\sum x_i P_i$	$\int x f(x) dx$
Second Moment $E[X^2]$	$\sum x_i^2 P_i$	$\int x^2 f(x) dx$
Total Probability	$\sum P_i = 1$	$\int f(x) dx = 1$

3. Case Study: Gambling Scenario and Expected Value

This example demonstrates why the probability of winning can be misleading compared to expected value.

3.1 Scenario Parameters

Winning Conditions:

- Roll 2, 3, or 4 → Win ₹1
- Roll 5 → Win ₹2

Losing Conditions:

- Roll 1 or 6 → Lose ₹3 (-3)

3.2 Probability vs Expectation Analysis

Probability of winning = $2/3$

However, expected value per game is negative.

Table 2: Payoff Matrix

Outcome	Probability (P_i)	Value (x_i)	$x_i P_i$
2, 3, 4	$3/6 = 1/2$	+1	$+1/2$
5	$1/6$	+2	$+2/6$
1, 6	$2/6 = 1/3$	-3	-1
TOTAL	1	-	$-1/6$

Conclusion:

Expected value per game = $-1/6$

Over n trials:

Total result = $n \times (-1/6)$

Since expectation is negative, long-term loss is guaranteed.

4. Measures of Dispersion

Knowing the center is insufficient; dispersion must also be measured.

4.1 Variance ($\text{Var}(X)$)

Variance measures deviation from the mean.

Formula: $\text{Var}(X) = E[X^2] - (E[X])^2$

Property: Variance is always positive.

4.2 Standard Deviation (σ)

Standard deviation is the square root of variance.

Formula: $\sigma = \sqrt{\text{Var}(X)}$

Physics analogy: Root Mean Square (RMS)

5. Applied Integration Examples

5.1 Pitch Diameter Function

Given:

$f(x) = \pi(1 + x^2) / 4$
for $x \in [0, 1]$

Expected value: $E[X] = \int_0^1 x f(x) dx$

Using substitution ($1 + x^2 = t$):

Result: $\pi / (2 \ln 2)$

5.2 Exponential Distribution Example

For functions involving $e^{(-\lambda x)}$:

Mean:

$$E[X] = 1 / \lambda$$

Second Moment:

$$E[X^2] = 2 / \lambda^2$$

Variance:

$$\text{Var}(X) = (2 / \lambda^2) - (1 / \lambda)^2 = 1 / \lambda^2$$